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RPA Based Insurance Claims Intake & Validation Automation System

RPA BUSINESS ANALYSIS & PROJECT MANAGEMENT FINAL CASE STUDY REPORT

1. Executive Summary

This case study explores the design of an RPA-based solution to automate the intake and validation of health insurance claims. In many insurance organizations, the initial stages of claim processing are still handled manually, requiring claims officers to review submissions, verify policy details, check supporting documents, and route claims for further assessment. These activities are repetitive, time-consuming, and prone to delays and human error.

The proposed solution introduces Robotic Process Automation (RPA) to streamline the early-stage claims workflow. The automation focuses on capturing claim submissions, extracting claim data, validating policy information, checking document completeness, classifying claims, and routing them to the appropriate processing queues. By automating these rule-based tasks, the organization can reduce manual effort, improve validation consistency, and accelerate claim intake processing.

This case study was developed from a Business Analysis and Product Management perspective, covering problem identification, requirements definition, automation design, backlog planning, testing scenarios, and expected business value.

2. Business Context

Health insurance providers receive claims from policyholders, hospitals, and healthcare providers. Each claim typically includes claim forms, medical invoices, policy details, and supporting medical documents. Before a claim can be assessed, the insurer must first validate whether the claim submission is complete and whether the policy is eligible for coverage.

In a manual process, claims officers perform several repetitive tasks such as reviewing documents, checking policy details, entering data into systems, and routing claims to the correct team. As claim volumes increase, this creates operational

bottlenecks, longer turnaround times, and higher administrative workload. This makes claims intake and validation a strong candidate for RPA implementation.

3. Problem Statement

The current health insurance claims intake process is highly manual and inefficient. Claims officers must manually review claim submissions, verify policy details, check document completeness, and determine whether a claim is valid, incomplete, or requires escalation. This results in:

- Delays in processing incoming claims
- High manual effort for repetitive validation tasks
- Increased risk of data entry and routing errors
- Inconsistent validation outcomes
- Reduced operational efficiency during high claim volumes

There is a need for a rule-based automation solution that can handle repetitive intake and validation activities while allowing claims staff to focus on exceptions and higher-value work.

4. Project Objectives

The key objectives of this project are:

- Automate the intake of health insurance claims from digital submission channels
- Validate policy information against internal systems
- Check whether required supporting documents are attached

- Identify incomplete or invalid claims early in the process
- Classify claims into valid, incomplete, or exception categories
- Route claims to the appropriate processing queue
- Reduce manual effort and improve processing speed
- Improve consistency and accuracy in claims validation

5. Scope

In Scope

- Claim intake from email or digital submission sources
- Extraction of key claim information
- Policy validation
- Mandatory document verification
- Claim classification
- Queue-based claim routing
- Exception logging and notification handling

Out of Scope

- Final claim approval or rejection decisions
- Fraud detection and investigation
- Payment or reimbursement processing
- Medical assessment and clinical decision-making
- Advanced AI-based claim adjudication

6. Current State Overview

In the current process, claims are received through email, branch channels, or digital forms. Claims intake officers manually open the claim, review attached documents, verify policy details in internal systems, enter claim information into the claims platform, and determine whether the claim is complete. If documents are missing or the policy is invalid, the officer must manually follow up or route the claim for further review.

This process is repetitive and highly dependent on human effort. It also introduces the risk of inconsistency, especially when multiple officers handle claims under different workloads or interpretation of validation rules.

7. Proposed Future-State Solution

The proposed future-state solution introduces an RPA bot that acts as a digital worker during the claim intake stage.

The bot will:

1. Capture incoming claims from email or portal submissions
2. Extract key claim details such as policy number, claimant details, claim type, and incident date
3. Validate the policy number and policy status against internal systems
4. Check whether all mandatory supporting documents are attached
5. Apply predefined business rules to assess claim completeness
6. Classify the claim as **Valid**, **Incomplete**, or **Exception**
7. Route the claim to the appropriate processing queue
8. Log processing outcomes and trigger notifications where required

This future-state design reduces manual workload while improving speed, consistency, and visibility in the intake process.

8. RPA Solution Summary

The automation solution is designed around a rule-based workflow. The bot is triggered whenever a new claim is received and performs a series of structured validation checks before routing the claim.

Core Automation Functions

- Claim data capture and extraction
- Policy existence and status validation
- Mandatory document verification
- Duplicate/basic validation checks
- Claim classification
- Queue-based routing
- Exception logging
- Notification support for incomplete claims

Output Queues

- **Valid Claims Queue** – for claims ready for further assessment
- **Incomplete Claims Queue** – for claims missing documents or mandatory information
- **Exception Review Queue** – for invalid policies, duplicates, unreadable files, or system issues

9. Requirements Summary

The key requirements identified for the solution include:

Business Requirements

- Automate claim intake and validation activities
- Improve turnaround time for claim registration
- Reduce manual effort and processing errors
- Standardized validation and routing decisions

Functional Requirements

- Capture and extract claim details from incoming submissions
- Validate policy number and active status
- Check claim completeness and required documents
- Classify claims and route them to the correct queue
- Generate notifications for incomplete claims
- Maintain audit logs for all bot actions

Non-Functional Requirements

- Secure handling of policyholder and medical data
- High validation, accuracy and reliability
- Auditability and traceability of all actions
- Scalability to handle increasing claim volumes

10. Business Rules Summary

The automation uses a set of predefined business rules to validate incoming claims. These include:

- Policy number must exist in the system
- Policy must be active on the incident date
- Claim type must be covered under the policy
- Mandatory claim fields must be completed
- Required supporting documents must be attached
- Potential duplicate claims should be flagged
- Invalid or incomplete claims must be routed to the correct queue

These rules form the decision logic used by the bot to determine claim status and routing.

11. Testing Summary

Testing was designed to validate the automation from a business and process perspective. The test coverage focused on common intake scenarios and expected routing outcomes.

Example scenarios covered:

- Valid claim with complete documents
- Claim with missing mandatory document
- Invalid policy number
- Expired or inactive policy
- Duplicate claim submission
- Corrupted or unreadable attachment
- Notification for incomplete claims
- Audit logging of processed claims

The objective of testing is to ensure that the bot consistently performs validation, classification, routing, and exception handling according to defined business rules.

12. Expected Business Value

The proposed automation is expected to deliver both operational and service-level benefits.

Operational Benefits

- Reduced manual effort in claim intake and validation
- Faster processing of incoming claims
- Reduced backlog at the intake stage
- Improved consistency in validation and routing

Quality Benefits

- Lower risk of manual data entry errors
- Standardized validation against business rules
- Better audit visibility and traceability

Customer / Service Benefits

- Faster acknowledgement of submitted claims
- Improved turnaround time for claim intake processing
- Better communication for incomplete claim submissions

13. Risks and Considerations

While the automation provides strong value, there are several considerations for implementation:

- Poor-quality or unstructured documents may affect extraction accuracy
- Frequent changes to claim rules may require bot updates
- Some claims will still require manual review due to exceptions
- Integration reliability depends on access to policy and claims systems
- Sensitive health and policy data must be handled securely

These risks can be managed through clear business rules, controlled bot access, exception queues, audit logging, and phased rollout.

14. Conclusion

This case study demonstrates how RPA can be applied to improve the health insurance claims intake and validation process by automating repetitive, rules-based tasks that currently consume significant manual effort. By introducing automation into claim capture, policy verification, document checks, and routing, the proposed solution helps reduce operational inefficiencies while improving speed, consistency, and process visibility.

From a Business Analysis and Product Management perspective, the project highlights how automation opportunities can be identified, structured into requirements, translated into a future-state solution, and validated through testing and phased delivery planning. The outcome is a practical and scalable automation concept that supports better claims operations while allowing human teams to focus on exceptions and higher-value decision-making.